

WARMUP: BREAK YOUR DAY 2 PAGE ON PURPOSE (LEVEL 1, ~15 MIN)

Open the interactive page you built yesterday ([greet.html](#) + [greet.js](#), the survey, or whatever you finished). Make a copy of both files first so you don't lose your working version. Then, one break at a time, save, refresh, and observe:

Break 1: In your JS, change `document.getElementById("name-input")` to `document.getElementById("nam-input")` (typo). What error appears in the console? Where? `greet.js:7 Uncaught TypeError: Cannot read properties of null (reading 'value')` at `HTMLFormElement.<anonymous>` (`greet.js:7:26`), On line 7, the form. So I would have to look at anything related to the form, the variables for example.

Break 2: Restore Break 1. Now comment out `event.preventDefault();`. Submit the form. What happens? Why does the result disappear? The error disappears, because the page refreshes.

Break 3: Restore Break 2. Now change `nameInput.value` to `nameInput.values` (plural). Click submit. What does the greeting show? Is there an error? Greeting displays nothing.

`greet.js:7 Uncaught ReferenceError: nameInputs is not defined at HTMLFormElement.<anonymous> (greet.js:7:16)`

Break 4: Restore Break 3. Now move `<script src="greet.js"></script>` from the bottom of `<body>` to inside `<head>`. Save, refresh, click submit. What happens? Why? Because it is reading an event listener for a form that hasn't been read yet. Everything must go in sequence.

`greet.js:5 Uncaught TypeError: Cannot read properties of null (reading 'addEventListener')` at `greet.js:5:6`

EXERCISE 1: READ THE ERROR (LEVEL 1, 25 MIN)

Below are 5 error messages. For each one, answer:

A.

1. What TYPE of error is it? A reference error, it used a variable that doesn't exist.

2. What does the message mean in plain English? Look at line three, line 3 says greeting, look at greeting and username is spelled incorrectly, it should be userName.
3. What line is the error on? Line 3 points to line 2
4. How would you fix it? You change the username to userName.

B.

1. What TYPE of error is it? A type error on line 2.
2. What does the message mean in plain English? Fruits is declared but has not been assigned a value, undefined has no .length property to read
3. What line is the error on? Line 2
4. How would you fix it? Intialize the variable first: let fruits = []; or let fruits = ["apple", "banana"];

C.

1. What TYPE of error is it? Is a syntaxx error,
2. What does the message mean in plain English? A syntaxx error is a missing bracket, comma or unclosed string for example.
3. What line is the error on? Line 4
4. How would you fix it? Remove the bracket.

D.

1. What TYPE of error is it? It is `TypeError: document.getElementById`
2. What does the message mean in plain English? It means look at the element ID to find the error.
3. What line is the error on? Line 1
4. How would you fix it? Change the ID to Id

E.

1. What TYPE of error is it? `Uncaught SyntaxError: Unterminated string literal`
2. What does the message mean in plain English? It is an error that points to the string.
3. What line is the error on? The error is on line 1
4. How would you fix it? Complete the string with a quote at the end of the string.

EXERCISE 2: CONSOLE.LOG DETECTIVE (LEVEL 1, 30 MIN)

This function is supposed to calculate the average of three test scores. It runs without errors, but it gives the wrong answer.

Step 1: Read the code. What SHOULD the answer be for scores of 80, 90, and 70? The bug is on the line `const average = sum / 2;` it should divide by 3 because there are three scores, not two.

Step 2:

Variable, Expected Value, Actual Value, Correct.

Score1, 80, 80, yes

Score2, 90, 90, yes

Score3 70, 70, yes

Sum, 240, 240, yes

Average, 80, 120, no (incorrect)

Step 4: Which line has the bug? Fix it. Run the code again. Does the output match the expected result? Yes

Yes the bug is the math. It should be `const discounted = price * 0.90;`

Yes — after the fix, `applyDiscount(150)` returns 135 and `applyDiscount(80)` still returns 80, both matching the expected output.

EXERCISE 3: DEVTOOLS SCAVENGER HUNT (LEVEL 2, 30 MIN)

Open this page in your browser: take any HTML page you've built this week (your about-me page, your cards page, etc.) and open DevTools.

Challenge 1: Console Tab

Type `2 + 2` and press Enter. What do you see? 4

Type `document.title` and press enter, What does it say when you return? It says 'About me'

Type `document.title = "I changed the title!"` and press Enter. Look at the browser tab. What happened? It changed the title in the tab of the browser.

Challenge 2: Elements tab:

Double-click the color value and change it to `red`. What happens? It turned the font to red

Find a hidden challenge: add a new CSS property right in DevTools. Add `text-decoration: underline;` to the heading. Did it work? Yes it underlined my text in red

Challenge 3 -- Box Model:

Find the box model diagram. Write down the margin, border, padding, and content dimensions for that element. Padding 64, 20; Content: 720 X 1081.590; Border: 0; Margin 0, 62,500

Challenge 4 -- Network tab:

How many files loaded? List them.: greet.html, greet.js, style.css, and photo file.
If you have an image on your page, find it in the Network tab. How large is the file?
content-length: 89342 bytes
If your CSS is in an external file, find it in the Network tab. Click on it. Can you see your CSS? Yes under preview under network tab.

Challenge 5 -- Live editing:

Change the text to something funny

Refresh the page. The changes disappear. Why? Because you are only changing it in the browser. Once the browser is refreshed, all the changes clear.

EXERCISE 4: FIX AI'S BUGS (LEVEL 2, 35 MIN)

AI generated these 4 small programs. Each one has exactly ONE bug. Find and fix each bug WITHOUT asking AI for help first. After you fix it, you can ask AI to verify your fix.

Buggy Program 1: Greeting Generator index.html:19 Uncaught TypeError: Cannot set properties of null (setting 'textContent') at index.html:19:54, Line 19 Id is missing an i.

Buggy Program 2: Color Changer: No console error, The text doesn't change colors when you click the button. The error was on line 12 colours instead of color.

Buggy Program 3: Counter

```
textContent = count (count was in a string) Corrected the error to this document.getElementById("count").textContent = count;
```

Buggy Program 4: Temperature Message

EXERCISE 5: DESCRIBE A BUG TO AI (LEVEL 2, 20 MIN)

Pick one of the bugs from Exercise 4 (before you fixed it). Write a bug description using the format from lecture:

1. Did AI identify the correct bug? Yes, immediately
2. Was AI's suggested fix correct? Yes it showed me the correct code.
3. Was AI's fix the same as yours, or different? AI fixed the same this I fixed.
4. Did AI explain WHY the bug happened? `document.getElementById("count").textContent = count;`
Yes, AI explained: "count" (with quotes) → just text ; count (no quotes) → your variable (the number increasing)

EXERCISE 6: DEBUG A MULTI-FILE PROJECT (LEVEL 3, 30 MIN)

Here's a small project with three files. There are 2 bugs spread across these files. Find and fix both. No hints -- part of the challenge is figuring out which files have bugs and which don't. Not every file has a bug; deciding "this file is fine" is part of real debugging.

The errors I found was In CSS alignitems has a missing dash. It should be align-items. I named my files main.js and style.css and learned in html that they didn't match so I fixed that. Then, I used AI for the last error. The last error was in html: `<ul id="task-list"></ol>` instead of `<ul id="task-list"></ul>`